

GenAI. Lecture 5 - Alpha-GAN, AAE

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LAMBDA • HSE

Recap. VAE



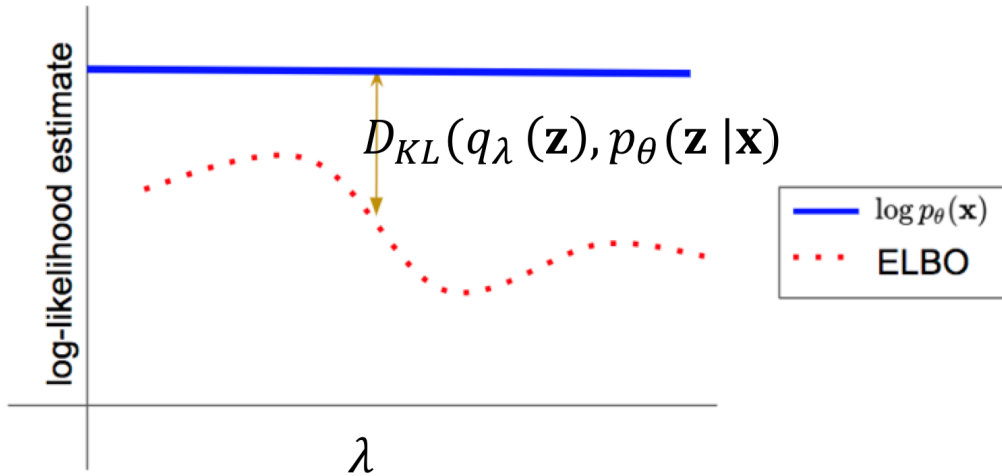
VAE Objective Function

The Variational Autoencoder (VAE) maximizes the Evidence Lower Bound (ELBO) to approximate the true data likelihood:

$$\log p(x) = \mathbb{E}_{z \sim q}[\log p(x|z)] + KL(q(z|x) \| p(z|x)) - KL(q(z|x) \| p(z))$$

- ▶ The first term is the expected log-likelihood, promoting accurate reconstruction.
- ▶ The second term ensures the approximated posterior $q(z|x)$ is close to the real posterior $p(z|x)$ and can not be estimated directly.
- ▶ The third term penalizes divergence between q and $p(z)$.
- ▶ The **first** and **third** terms are **ELBO**

ELBO



VAE Loss Function

- ▶ The VAE loss function has two components:
 1. **Reconstruction Loss:** Measures how well the decoder reconstructs the input.

$$\mathcal{L}_{recon} = \mathbb{E}_{q(z|x)}[\log p(x|z)]$$

2. **KL Divergence:** Regularizes the latent distribution to match the prior distribution (usually standard normal).

$$\mathcal{L}_{KL} = D_{KL}(q(z|x)||p(z))$$

- ▶ Total Loss:

$$\mathcal{L} = \mathcal{L}_{recon} + \mathcal{L}_{KL}$$

KL Divergence Term

- ▶ The KL Divergence measures the difference between the approximate posterior $q(z|x)$ and the prior $p(z)$.
- ▶ For a Gaussian prior $p(z) = \mathcal{N}(0, I)$ and posterior $q(z|x) = \mathcal{N}(\mu, \sigma^2)$, the KL term simplifies to:

$$D_{KL}(\mathcal{N}(\mu, \sigma^2) \parallel \mathcal{N}(0, I)) = \frac{1}{2} \sum (\sigma^2 + \mu^2 - 1 - \log \sigma^2)$$

- ▶ This encourages the latent variables to follow a standard normal distribution.

Adversarial AutoEncoder (AAE)



KL Divergence in AAE

- ▶ In Adversarial Autoencoders, the KL divergence term is replaced by a discriminator-based loss.
- ▶ The goal is to match the aggregated posterior $q(z)$ with the prior $p(z)$.

$$KL(q(z|x)||p(z)) \rightarrow BCE(y \sim \mathcal{D}, z \sim p(z))$$

- ▶ Here, BCE refers to the Binary Cross-Entropy loss used to train the discriminator $D(z)$.
- ▶ The discriminator distinguishes between samples from the prior $p(z)$ and the encoder output $q(z|x)$.

Discriminator Loss in AAE

- ▶ The discriminator $D(z)$ is trained to maximize:

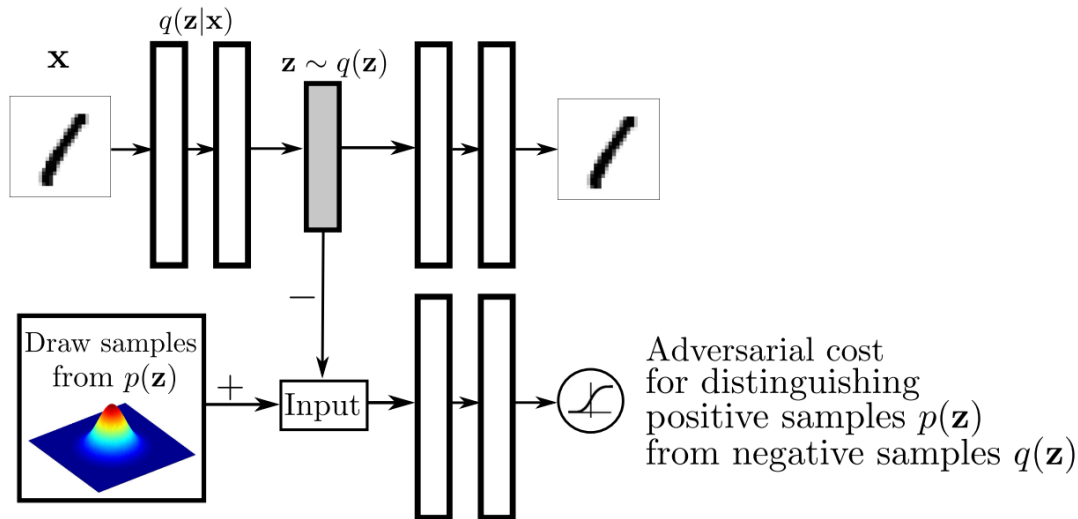
$$\mathbb{E}_{z \sim p(z)}[\log D(z)] + \mathbb{E}_{z \sim q(z|x)}[\log(1 - D(z))]$$

- ▶ The encoder is trained to fool the discriminator by minimizing:

$$\mathbb{E}_{z \sim q(z|x)}[\log D(z)]$$

- ▶ This adversarial training aligns the latent distribution $q(z)$ with the prior $p(z)$.

AAE architecture



α -GAN



Optimal Bayesian Classifier for Density Ratio Estimation

- ▶ The optimal discriminator $D^*(z)$ in Alpha-GAN corresponds to the Bayes optimal classifier.
- ▶ It estimates the probability of z being sampled from the prior $p(z)$ versus the approximate posterior $q(z|x)$.

$$D^*(z) = \frac{p(z)}{p(z) + q(z|x)}$$

- ▶ The density ratio can be derived from this optimal classifier:

$$\frac{q(z|x)}{p(z)} = \frac{1 - D^*(z)}{D^*(z)}$$

- ▶ This approach enables efficient and accurate estimation of complex posterior distributions.

Density Ratio Estimation in Alpha-GAN

- ▶ The KL divergence term in traditional VAEs is replaced using the Density Ratio Estimation (DRE) trick.
- ▶ A discriminator $D(z)$ is used to approximate the density ratio between the approximate posterior $q(z|x)$ and the prior $p(z)$.

$$KL(q(z|x)|p(z)) \approx \mathbb{E}_{z \sim q(z|x)} \log \frac{D(z)}{1 - D(z)}$$

- ▶ This allows for flexible matching of complex distributions without explicit KL divergence computation.

Final loss

$$\log p(x) \geq$$

$$\mathbb{E}_{z \sim q}[\log p(x|z)] - KL(q(z|x) \| p(z)) = \mathbb{E}_{z \sim q}[\log p(x|z)] - \mathbb{E}_{z \sim q(z|x)} \log \frac{D(z)}{1-D(z)} =$$

$$\mathbb{E}_{z \sim q}[\log p(x|z)] - \mathbb{E}_{z \sim q}[\log p(x)] - \mathbb{E}_{z \sim q(z|x)} \log \frac{D(z)}{1-D(z)} + \mathbb{E}_{z \sim q}[\log p(x)] =$$

$$\mathbb{E}_{x \sim p(x|z)} \log \frac{p(x|z)}{p(x)} - \mathbb{E}_{z \sim q(z|x)} \log \frac{D(z)}{1-D(z)} + \text{const} \approx$$

$$\alpha \mathbb{E}_{x \sim p(x|z)} \log \frac{D_x(x)}{1-D_x(x)} + (1 - \alpha) MSE(x, x_{\text{recon}}) - \mathbb{E}_{z \sim q(z|x)} \log \frac{D_z(z)}{1-D_z(z)}$$